

1. Overdraw and Overshading (Task 1.1)

The visualisation modes are implemented using specialised graphics pipelines with Additive Blending (VK_BLEND_OP_ADD, ONE, ONE).

- Shader: Both modes reuse `overdraw.frag`, which outputs a constant colour value (0.05 per channel) and alpha 0.0 (unused).
- Control Keys:
 - Key 6: Overdraw (Depth Test OFF). This accumulates colour for every fragment generated by the geometry, regardless of occlusion. It visualises the total Depth Complexity of the scene. Brighter areas indicate many layers of geometry (visible or hidden) are being rasterized.
 - Key 7: Overshading (Depth Test ON). This accumulates colour for every fragment that passes the depth test. It visualises Shading Complexity (redundant fragment shader invocations) caused by the draw order. If objects are drawn Back-to-Front, the pixel is shaded multiple times (bright). If Front-to-Back, it is shaded once (dark).

Overdraw: Depth Complexity (see Figure 2)

- Looking down the row of columns from either side produces a Symmetric result. The number of overlapping triangles (depth complexity) is constant regardless of the viewing angle or draw order.
- High overdraw is observed in the foliage and when looking through the aligned columns, appearing nearly white due to 20+ layers of accumulated geometry.

Overshading: Shading Cost (see Figure 3)

- Looking down the columns from one side vs. the other produces a roughly Symmetric result (similar brightness/cost).
- This symmetry key follows expectation for an unsorted scene.
- Since the renderer does not implement dynamic sorting, the index buffer order is static. The geometry is not optimised for any specific viewpoint. Therefore, from either end, the GPU encounters a random mix of "Front-to-Back" (efficient) and "Back-to-Front" (inefficient) triangles, resulting in an average overshading cost that is consistent across viewpoints. An Asymmetric result would only be expected if the objects were stored in a strictly linear order in memory.

Beyond identifying optimisation targets, these modes are critical for

- Identifying worst-case camera angles where frame rate might drop.
- Detecting hidden heavy geometry (like objects inside a treasure chest) that shouldn't be rendered.
- Distinguishing whether a bottleneck is Vertex-bound or Fragment-bound.

Unimplemented optimisation

- Baseline Overdraw: Reduce by visibility culling (frustum/occlusion) to send fewer triangles.
- Basic Overshading: Reduce by sorting objects Front-to-Back or using a Z-Prepass to prime the depth buffer.



Figure 1. Standard shaded rendering. Views from opposite sides of the column row are shown side by side.



Figure 2. Overdraw visualisation (depth complexity). Views from opposite sides of the column row are shown side by side and appear symmetric.



Figure 3. Overshading visualisation (shading cost). Views from opposite sides of the column row are shown side by side and appear symmetric.

2. Shadow Mapping (Task 1.5)

A dedicated shadow rendering pass renders the scene depth from the light's point of view into a 2048x2048 depth texture (D32_SFLOAT).

Pipeline

- Vertex Shader: Transforms geometry using LightVP.
- Fragment Shader: Discards fragments based on the Alpha Mask (for transparent foliage shadows). The pipeline has colour writes disabled, so valid fragments only update the Depth Buffer.
- State: Standard Back-Face Culling is used (Front-Face Culling is disabled). Instead, Depth Bias is applied to solve shadow acne.

Parameters

- Resolution: 2048x2048. Chosen as the functional minimum for this scene. Lower resolutions like 512 and 1024 failed to resolve scene geometry correctly, resulting in broken lighting and missing shadows.
- Projection: Perspective Projection (90-degree FOV). While sunlight is physically directional (Orthographic), a wide-angle Perspective projection was chosen here as a practical scene-specific optimisation. It effectively covers the bounded volume of the ruins. This maximises the usage of the shadow map texels for the geometry closest to the player, simulating a localised key light.
- Bias: `vkCmdSetDepthBias(1.25f, 0.0f, 1.75f)`. A combination of Constant Bias (1.25) and Slope Factor (1.75) was tuned to remove shadow acne on the floor without creating detached shadows on the statue base.
- Filtering: A 3x3 Percentage-Closer Filtering (PCF) kernel is implemented in `default.frag`. It takes 9 samples around the projected coordinate and averages the comparison results.



Figure 4. Left: without PCF, hard and aliased shadows. Right: with 3x3 PCF, softened shadow edges.

Results & Comparison

- Without PCF: Shadows are hard and aliased. The boundaries show pixelation corresponding to the shadow map texels (see Figure 4).
- With 3x3 PCF: Shadow edges are softened, masking the underlying resolution aliasing and providing a more natural look (see Figure 4).
- Resolution Impact:
 - 512x512: Functional failure (see Figure 5). The resolution was too low to resolve valid depth comparisons, resulting in missing highlights ("no shining light") and no visible shadows. This is likely due to all fragments failing the depth test or extreme bias interaction.
 - 1024x1024: Partial functionality (see Figure 6). Some highlights appear, but shadows are inconsistent or incorrect, suggesting the shadow map precision was still insufficient for the scene scale.
 - 2048x2048: Correct and stable (see Figure 7). This resolution marks the threshold where the shadow map successfully resolves the scene geometry, producing accurate highlights and detailed, sharp shadows.
 - 4096x4096: Aborted. The trial was terminated because the increase in resource usage caused the application to stall during startup, so it was terminated.



Figure 5. Visualisation of the 512x512 Shadow Map.



Figure 6. Visualisation of the 1024x1024 Shadow Map.



Figure 7. Visualisation of the 2048x2048 Shadow Map.

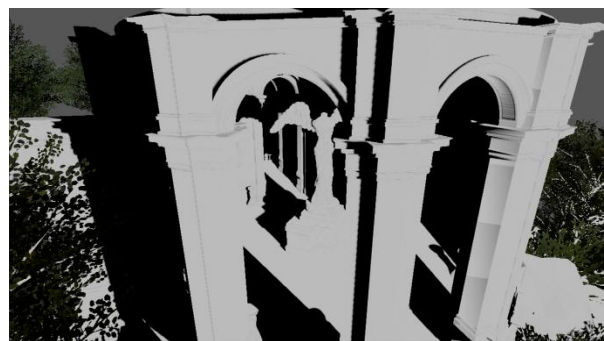


Figure 8. Visualisation of the 2048x2048 Shadow Map (Depth Buffer) from the Light's perspective (key 8).

3. References

- [1] Rafael Kuffner Dos Anjos and Markus Billeter (2025) COMP5892M Advanced Rendering: Assignment 2 Coursework Brief. University of Leeds.
- [2] Rafael Kuffner Dos Anjos and Markus Billeter (2025) *Advanced Rendering Lecture Slides*. COMP5892M Advanced Rendering, University of Leeds.